

STAT 516 – Exam 1 Answer Key

Summer 2026

1. **[15 points]** A research team has 8 statisticians, 6 computer scientists, and 5 engineers. A committee of 7 people is selected at random.

Find the probability that the committee contains at least one person from each group and that no group has more than 3 people on the committee.

The total number of possible committees is

$$\binom{19}{7}.$$

We need at least one person from each group and no group has more than 3 people. The possible group-size patterns are

$$(1, 3, 3), (3, 1, 3), (3, 3, 1), (2, 2, 3), (2, 3, 2), (3, 2, 2),$$

where the entries correspond to statisticians, computer scientists, and engineers.

Thus,

$$P = \frac{\binom{8}{1}\binom{6}{3}\binom{5}{3} + \binom{8}{3}\binom{6}{1}\binom{5}{3} + \binom{8}{3}\binom{6}{3}\binom{5}{1} + \binom{8}{2}\binom{6}{2}\binom{5}{3} + \binom{8}{2}\binom{6}{3}\binom{5}{2} + \binom{8}{3}\binom{6}{2}\binom{5}{2}}{\binom{19}{7}}.$$

Therefore,

$$P = \frac{28760}{50388} = \frac{7190}{12597} \approx 0.5708.$$

2. **[20 points]** A bag contains three biased coins. Coin 1 has probability 0.25 of landing heads, Coin 2 has probability 0.50 of landing heads, and Coin 3 has probability 0.75 of landing heads.

One coin is selected at random and tossed four times. Let X be the number of heads observed.

- (a) Find $P(X = 3)$.
(b) Given that $X = 3$, find the probability that Coin 3 was selected.

Let C_i be the event that Coin i was selected. Then

$$P(C_1) = P(C_2) = P(C_3) = \frac{1}{3}.$$

For part (a),

$$P(X = 3) = \sum_{i=1}^3 P(X = 3|C_i)P(C_i).$$

Therefore,

$$P(X = 3) = \frac{1}{3} \left[\binom{4}{3} (0.25)^3 (0.75) + \binom{4}{3} (0.50)^3 (0.50) + \binom{4}{3} (0.75)^3 (0.25) \right].$$

Thus,

$$P(X = 3) = 0.2396.$$

For part (b),

$$P(C_3|X = 3) = \frac{P(X = 3|C_3)P(C_3)}{P(X = 3)}.$$

So,

$$P(C_3|X = 3) = \frac{\binom{4}{3} (0.75)^3 (0.25) \cdot \frac{1}{3}}{0.2396} \approx 0.5870.$$

3. [10 points] Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} \frac{3}{8}x^2, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\text{Var}(3X^2 - 2).$$

$$\text{Var}(3X^2 - 2) = 9 \text{Var}(X^2).$$

Now,

$$E(X^2) = \int_0^2 x^2 \cdot \frac{3}{8}x^2 dx = \frac{3}{8} \int_0^2 x^4 dx = \frac{12}{5}.$$

Also,

$$E(X^4) = \int_0^2 x^4 \cdot \frac{3}{8}x^2 dx = \frac{3}{8} \int_0^2 x^6 dx = \frac{48}{7}.$$

Thus,

$$\text{Var}(X^2) = E(X^4) - [E(X^2)]^2 = \frac{48}{7} - \left(\frac{12}{5}\right)^2.$$

Therefore,

$$\text{Var}(X^2) = \frac{192}{175}.$$

Hence,

$$\text{Var}(3X^2 - 2) = 9 \cdot \frac{192}{175} = \frac{1728}{175} \approx 9.8743.$$

4. [15 points] Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} \frac{c}{x^5}, & x > 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) [7 points] Find the value of c .

(b) [8 points] Find $P(|X - 1| < \frac{1}{2})$.

Part (a), since $f_X(x)$ is a PDF,

$$\int_1^{\infty} \frac{c}{x^5} dx = 1.$$

Now,

$$\int_1^{\infty} x^{-5} dx = \left[-\frac{1}{4}x^{-4} \right]_1^{\infty} = \frac{1}{4}.$$

Therefore,

$$\frac{c}{4} = 1,$$

so

$$c = 4.$$

Part (b),

$$|X - 1| < \frac{1}{2}$$

Thus,

$$\frac{1}{2} < X < \frac{3}{2}.$$

Since the support is $x > 1$, the event becomes

$$1 < X < \frac{3}{2}.$$

Therefore,

$$P\left(|X - 1| < \frac{1}{2}\right) = \int_1^{3/2} \frac{4}{x^5} dx.$$

$$P\left(|X - 1| < \frac{1}{2}\right) = [-x^{-4}]_1^{3/2}.$$

Thus,

$$P\left(|X - 1| < \frac{1}{2}\right) = 1 - \left(\frac{3}{2}\right)^{-4} = 1 - \left(\frac{2}{3}\right)^4 = \frac{65}{81}$$

5. [20 points] A random variable Y is generated using the following two-stage procedure. First, a machine is selected. Machine 1 is selected with probability 0.35, and Machine 2 is selected with probability 0.65. Second, after the machine is chosen, generate Y from that machine's density.

If Machine 1 is selected, then Y has density

$$f_1(y) = 2y, \quad 0 < y < 1.$$

If Machine 2 is selected, then Y has density

$$f_2(y) = 6y(1 - y), \quad 0 < y < 1.$$

(a) [12 points] Find $P(Y > 0.6)$.

(b) [8 points] Given that $Y > 0.6$, find the probability that Machine 2 was selected.

Let M_1 be the event that Machine 1 is selected, and let M_2 be the event that Machine 2 is selected.

Part (a), We are given

$$P(M_1) = 0.35, \quad P(M_2) = 0.65.$$

For Machine 1,

$$P(Y > 0.6|M_1) = \int_{0.6}^1 2y \, dy = [y^2]_{0.6}^1 = 1 - 0.36 = 0.64.$$

For Machine 2,

$$P(Y > 0.6|M_2) = \int_{0.6}^1 6y(1 - y) \, dy.$$

Since the CDF corresponding to $6y(1 - y)$ is

$$F(y) = 3y^2 - 2y^3,$$

we get

$$P(Y > 0.6|M_2) = 1 - [3(0.6)^2 - 2(0.6)^3] = 1 - 0.648 = 0.352.$$

$$P(Y > 0.6) = P(Y > 0.6|M_1)P(M_1) + P(Y > 0.6|M_2)P(M_2).$$

Thus,

$$P(Y > 0.6) = 0.64(0.35) + 0.352(0.65) = 0.4528.$$

For part (b),

$$P(M_2|Y > 0.6) = \frac{P(Y > 0.6|M_2)P(M_2)}{P(Y > 0.6)}.$$

Therefore,

$$P(M_2|Y > 0.6) = \frac{0.352(0.65)}{0.4528} \approx 0.5053.$$

6. [20 points] Let A , B , and C be three events such that

$$P(A) = 0.45, \quad P(B) = 0.50, \quad P(C) = 0.40,$$

$$P(A \cap B) = 0.25, \quad P(A \cap C) = 0.18, \quad P(B \cap C) = 0.20,$$

and

$$P(A \cup B \cup C) = 0.82.$$

(a) [15 points] Find $P(A \mid B \cup C)$.

(b) [5 points] Are A and $B \cup C$ independent? Justify your answer.

By inclusion-exclusion,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

Substitute the given values:

$$0.82 = 0.45 + 0.50 + 0.40 - 0.25 - 0.18 - 0.20 + P(A \cap B \cap C).$$

Thus,

$$P(A \cap B \cap C) = 0.10.$$

part (a),

$$P(A \mid B \cup C) = \frac{P(A \cap (B \cup C))}{P(B \cup C)}.$$

Now,

$$P(B \cup C) = P(B) + P(C) - P(B \cap C) = 0.50 + 0.40 - 0.20 = 0.70.$$

Also,

$$P(A \cap (B \cup C)) = P((A \cap B) \cup (A \cap C)).$$

Thus,

$$P(A \cap (B \cup C)) = P(A \cap B) + P(A \cap C) - P(A \cap B \cap C).$$

So,

$$P(A \cap (B \cup C)) = 0.25 + 0.18 - 0.10 = 0.33.$$

Therefore,

$$P(A \mid B \cup C) = \frac{0.33}{0.70} = \frac{33}{70} \approx 0.4714.$$

Part (b), A and $B \cup C$ are independent if

$$P(A \cap (B \cup C)) = P(A)P(B \cup C).$$

We have

$$P(A \cap (B \cup C)) = 0.33,$$

but

$$P(A)P(B \cup C) = 0.45(0.70) = 0.315.$$

Since

$$0.33 \neq 0.315,$$

A and $B \cup C$ are not independent.